

K-Means from Scratch in Python

Contents

1. Overview	2
2. Prerequisites	3
3. The shape of the class	3
4. Step 2: random starting centroids	4
5. Step 3: assigning clusters	5
5.1 One distance formula for any number of features	6
5.2 The nearest centroid	8
6. Step 4: moving the centroids	9
7. Step 5: stopping	10
8. Testing the class	11
9. Bad random starts	13
10. Summary	15
11. Sources	15
12. Key terms	15

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Unsupervised learning ([Note ML-003](#)); Feature scaling ([Note ML-006](#)); Elbow method and WCSS ([Note ML-122](#)); Expectation maximization (EM) ([Note MA-074](#)).
- **Leads to:** Hierarchical clustering ([Note ML-125](#)); DBSCAN ([Note ML-126](#)).
- **Compare with:** Hierarchical clustering ([Note ML-125](#)); DBSCAN ([Note ML-126](#)); Gaussian mixture model (GMM) ([Note MA-073](#)).

1. Overview

Key point: We write k-means (G-996) ourselves as a class with one public method, `fit_predict`, and two helpers: `assign_clusters` (step 3) and `move_centroids` (step 4).

Coding an algorithm from scratch is the surest test that we understand it. This Note turns the five steps of k-means (the [k-means Note](#), section 4) into about 40 lines of Python. Figure 1 shows how the steps map onto the methods of the class.

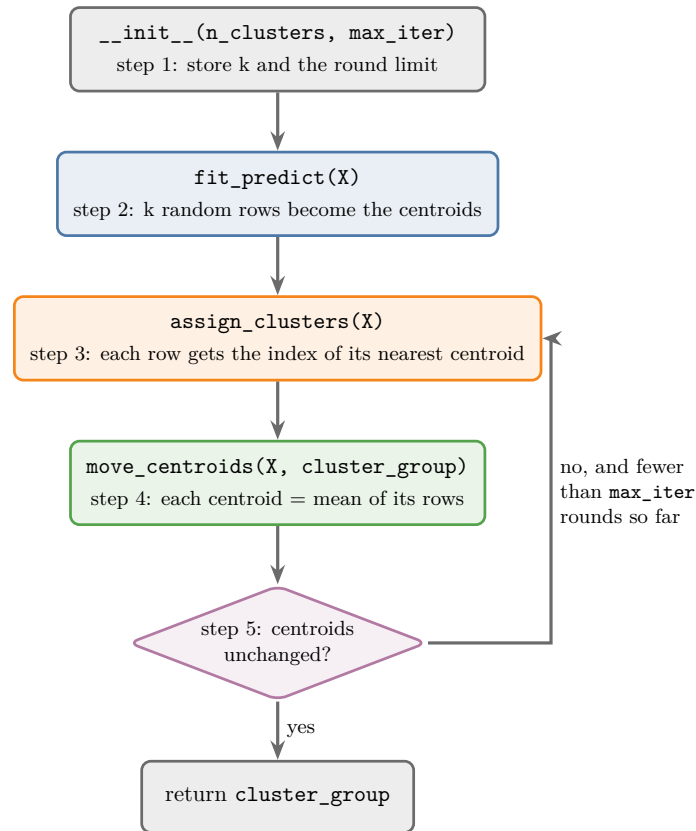


Figure 1: The KMeans class: which method does which step of k-means

The class is in `kmeans.py` in this Note's folder; running `python kmeans.py` checks that it finds the same clusters as scikit-learn. The Notebook (`notebook.ipynb`) runs every experiment below.

2. Prerequisites

- The five steps of k-means and WCSS (G-2102): the [k-means Note](#).
- KMeans in scikit-learn, the student data and boolean indexing: the [k-means in Python Note](#).

3. The shape of the class

Key point: The constructor stores `k` and a limit on the number of rounds; `fit_predict` runs the loop and returns one cluster number per observation.

We copy scikit-learn's interface, so our class is used exactly like the real one:

Python: Using the class.
`from kmeans import KMeans`

```
km = KMeans(n_clusters=4, max_iter=100)
y_means = km.fit_predict(X)    # one cluster number per row
```

from kmeans import KMeans loads the class from our own file `kmeans.py`, not from `scikit-learn`.

The **constructor** (G-457), the method that runs when the object is created, takes two settings:

- **n_clusters**: k, the number of clusters (step 1). Default 2.
- **max_iter**: the largest number of assign-and-move rounds. Default 100. The loop stops earlier if the centroids (G-367) stop moving.

Python: The constructor.

```
class KMeans:
    def __init__(self, n_clusters=2, max_iter=100,
                  random_state=None):
        self.n_clusters = n_clusters
        self.max_iter = max_iter
        self.random_state = random_state
        self.centroids = None    # set by fit_predict
```

`random_state` is our addition: it fixes the random start so that results can be repeated.

4. Step 2: random starting centroids

Key point: Pick k different row numbers at random and take those rows as the first centroids.

`X` is a 2-D NumPy array: with 100 points in 2 columns, it holds 100 rows of 2 numbers each. Each row is an **observation** (one record of the data) and each column a **feature** (an input variable). To start, we pick k different row numbers between 0 and 99 at random, then take those rows. Figure 2 shows one such start on the 200 students: the four chosen rows are real students. They fall in pairs inside two groups, and two groups start with no centroid at all; the loop of Figure 6 still recovers all four.

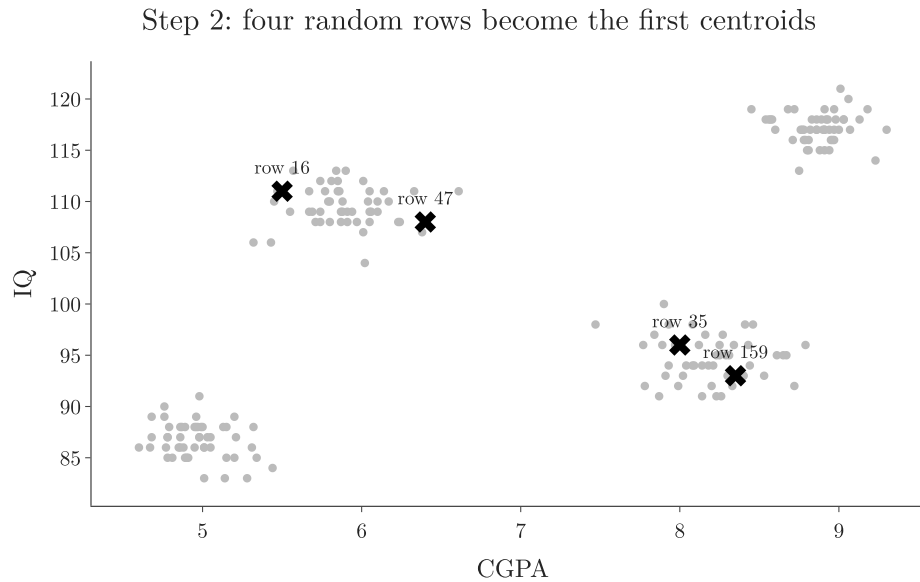


Figure 2: Step 2 on the students: rows 47, 16, 35 and 159, picked at random, become the first four centroids

Python: Choosing the starting centroids.

```
rng = np.random.default_rng(self.random_state)
random_index = rng.choice(X.shape[0],
                           size=self.n_clusters,
                           replace=False)
self.centroids = X[random_index].astype(float)
```

`X.shape[0]` is the number of rows. `rng.choice(100, size=2, replace=False)` returns 2 different numbers from 0 to 99, for example [26, 17]. `X[[26, 17]]` then picks rows 26 and 17. `astype(float)` lets the centroids hold decimals once they move.

Extra: The same can be written with Python's `random` module: `random.sample(range(X.shape[0]), k)` also returns `k` different numbers. NumPy's generator is used here because it takes a seed per object, so two models never share random state.

5. Step 3: assigning clusters

Key point: For every row, compute its distance to every centroid and record the index of the smallest one.

5.1 One distance formula for any number of features

Key point: The squared Euclidean distance (G-715) is the dot product (G-634) of the difference vector with itself, so `np.sqrt(np.dot(a - b, a - b))` works for 2, 3 or 100 features.

The Euclidean distance (the [KNN imputer Note](#), section 4.1) gains one term per feature, so code written term by term would change with the number of features. The sum of squared differences is the dot product (the [PCA step-by-step Note](#)) of the difference vector with itself, which needs no change:

$$d(a, b) = \sqrt{(b - a) \cdot (b - a)}$$

For $a = (1, 2)$ and $b = (4, 5)$:

$$b - a = (3, 3)$$

$$(3, 3) \cdot (3, 3) = 18$$

$$d = \sqrt{18} \approx 4.24$$

With a third feature, $(1, 2, 3)$ and $(4, 5, 6)$, the same code gives:

$$b - a = (3, 3, 3)$$

$$(3, 3, 3) \cdot (3, 3, 3) = 27$$

$$d = \sqrt{27} \approx 5.196$$

In Figure 3, watch the two dashed sides: each feature adds one squared side, and the dot product adds them all at once.

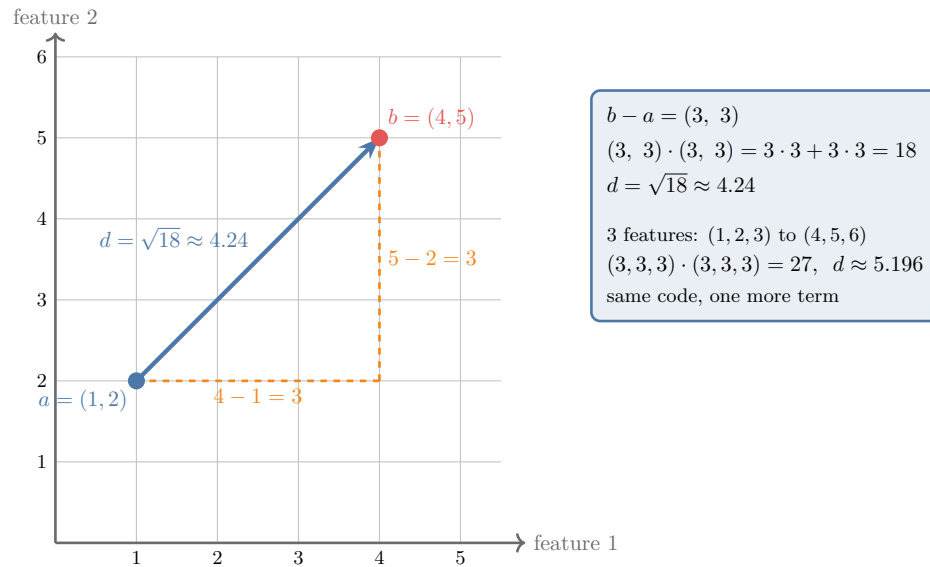


Figure 3: The distance from $(1, 2)$ to $(4, 5)$: the difference vector $(3, 3)$, its dot product with itself, and its square root

Figure 4 plays the same calculation step by step: the difference vector, its dot product with itself, the square root. Watch the last part: a third feature adds one more term to the dot product, and the line of code stays the same.

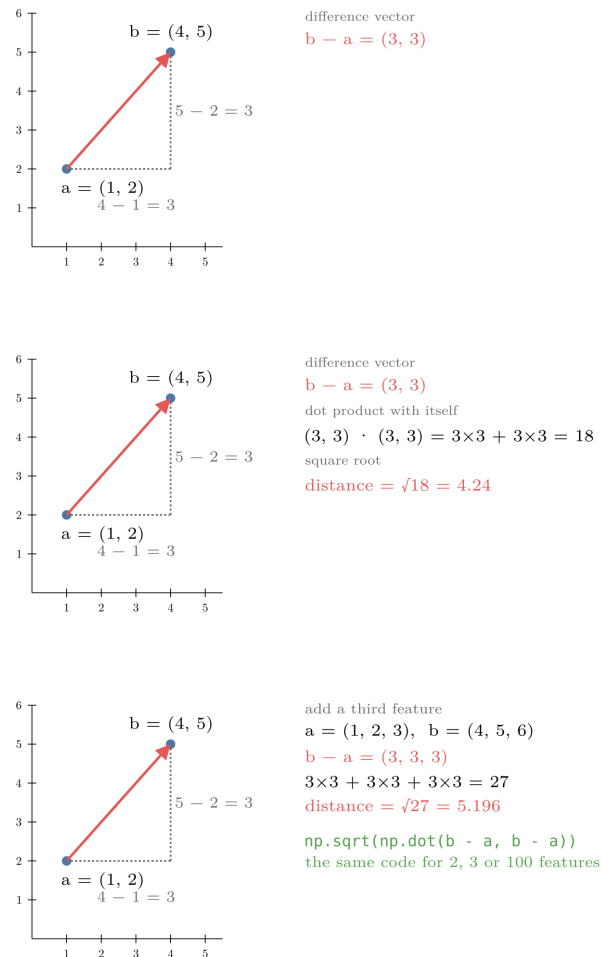


Figure 4: The distance from $(1, 2)$ to $(4, 5)$ as a dot product, step by step; then a third feature is added, $(1, 2, 3)$ to $(4, 5, 6)$, and the same code gives 5.196

Python: The distance in NumPy.

```
np.sqrt(np.dot(b - a, b - a))    # 4.24 for (1, 2), (4, 5)
```

$b - a$ subtracts element by element. `np.dot(u, v)` multiplies matching elements and adds them up.

5.2 The nearest centroid

Key point: With 100 rows and 2 centroids we compute 200 distances; each row gets the index of its smaller one.

A nested loop visits every row and, inside, every centroid. With 100 rows and 2 centroids the inner line runs:

$$100 \times 2 = 200$$

times. For each row we keep the position of the smallest distance: 0 if the first centroid is nearer, 1 if the second.

Python: `assign_clusters`.

```
def assign_clusters(self, X):
    cluster_group = []
    for row in X:
        distances = [np.sqrt(np.dot(row - c, row - c))
                     for c in self.centroids]
        cluster_group.append(int(np.argmin(distances)))
    return np.array(cluster_group)
```

The square brackets build a list of k distances, one per centroid. `np.argmin` returns the position of the smallest value, the same as `distances.index(min(distances))`. The result has one entry per row, such as `[0, 0, 1, ...]`, and is returned as a NumPy array so it can be used for boolean indexing.

6. Step 4: moving the centroids

Key point: For each cluster k , select its rows with `X[cluster_group == k]` and take the mean of each column with `mean(axis=0)`.

The cluster numbers tell us which rows belong to which cluster. The new centroid of a cluster is the mean of each column over its rows (the [k-means Note](#), section 4.3).

Take five rows and their clusters:

Row	Values	Cluster
0	(1, 2)	0
1	(2, 3)	1
2	(3, 4)	0
3	(4, 5)	0
4	(5, 6)	1

`X[cluster_group == 0]` keeps rows 0, 2 and 3. Their column means are:

$$(1 + 3 + 4)/3 = 2.67$$

$$(2 + 4 + 5)/3 = 3.67$$

So the new centroid of cluster 0 is (2.67, 3.67). Cluster 1 gets (3.5, 4.5). Figure 5 draws both: each cross lands in the middle of its own rows, and the other cluster's rows (grey) play no part.

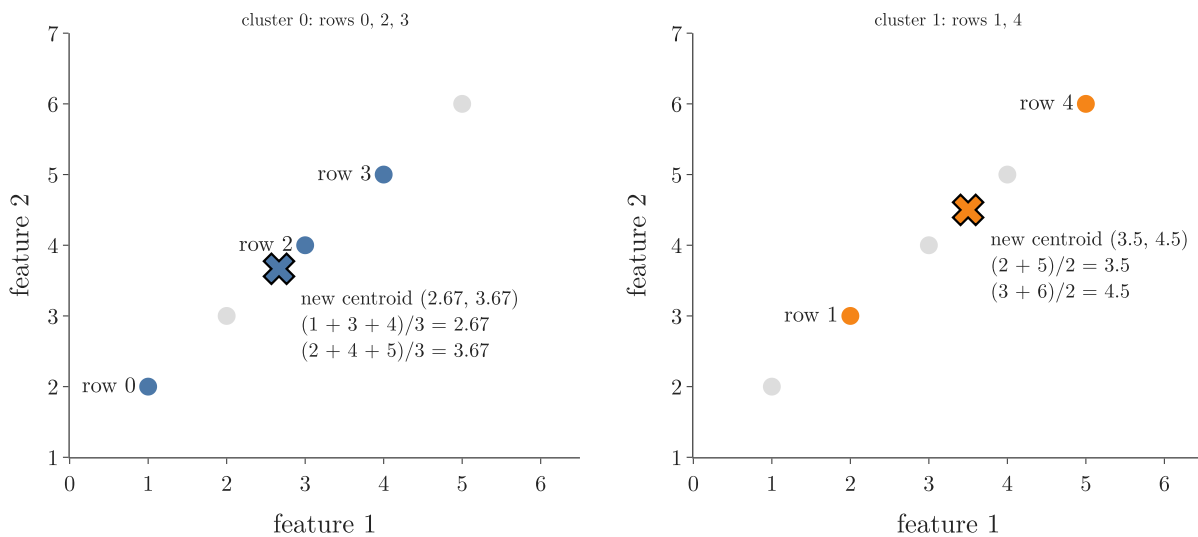


Figure 5: Moving the centroids for the five-row example: each new centroid is the column mean of its own rows

Python: `move_centroids`.

```
def move_centroids(self, X, cluster_group):
    new_centroids = []
    for k in range(self.n_clusters):
        members = X[cluster_group == k]
        new_centroids.append(members.mean(axis=0))
    return np.array(new_centroids)
```

`axis=0` averages down each column, giving one mean per column. Without it, `mean()` averages all six numbers into a single value (3.17), which is not a point at all.

Extra: A cluster can end up empty: no row is nearest to its centroid. The mean of zero rows is undefined, and a loop over the cluster numbers that actually occur (for example with `np.unique(cluster_group)`) would silently return fewer than `k` centroids. The file `kmeans.py` keeps an empty cluster's old centroid instead. scikit-learn moves it to the data point that is farthest from its own centroid (scikit-learn source, `_k_means_common.pyx`).

7. Step 5: stopping

Key point: Save the centroids before moving them; if the new ones equal the old ones, break out of the loop.

Python: The loop inside `fit_predict`.

```
for i in range(self.max_iter):
    cluster_group = self.assign_clusters(X)
```

```

old_centroids = self.centroids
self.centroids = self.move_centroids(X, cluster_group)
if np.allclose(old_centroids, self.centroids):
    break
return cluster_group

```

`break` leaves the loop at once. `np.allclose` is True when every pair of numbers is equal up to a tiny rounding error. `(old == new).all()` also works, but computed means can differ in the last decimal place.

Figure 6 runs the whole loop on the 200 students from the start of Figure 2. Watch the largest move shrink round by round; in round 4 the centroids do not move at all, `np.allclose` is True, and the loop breaks.

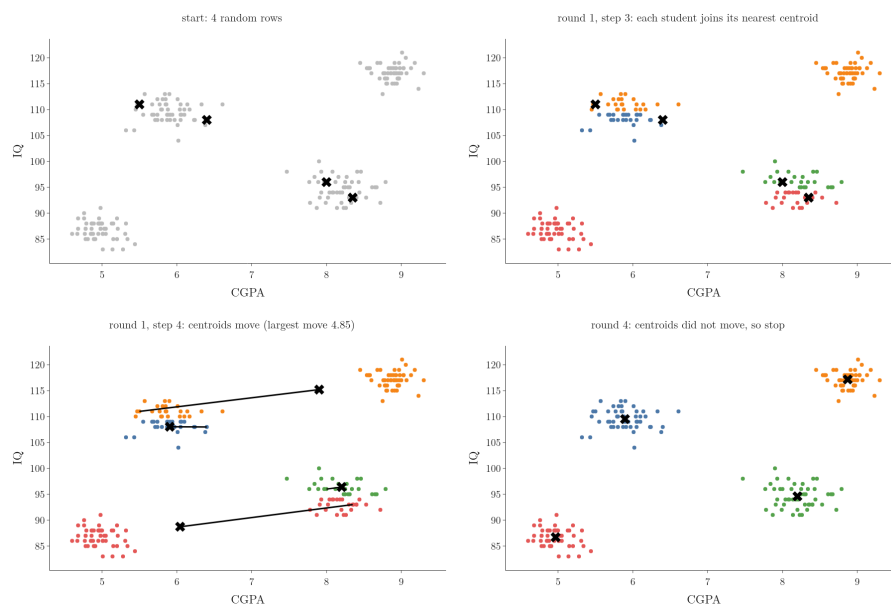


Figure 6: The assign-and-move loop on the students: colours change at step 3, crosses move at step 4, and the loop stops in round 4

The loop ends in one of two ways (Figure 1): the centroids stop moving, or `max_iter` rounds have run. Either way, `fit_predict` returns the cluster number of every row.

8. Testing the class

Key point: On two, three and four blobs, and on the 200 students, the class finds the right groups in 2 to 5 rounds.

We generate test data with `make_blobs` (the [k-means in Python Note](#), section 8): 100 points around 2, 3 or 4 centres, each with spread 1. Then we run our class with the matching `k`, and finally on the 200 students with `k = 4`.

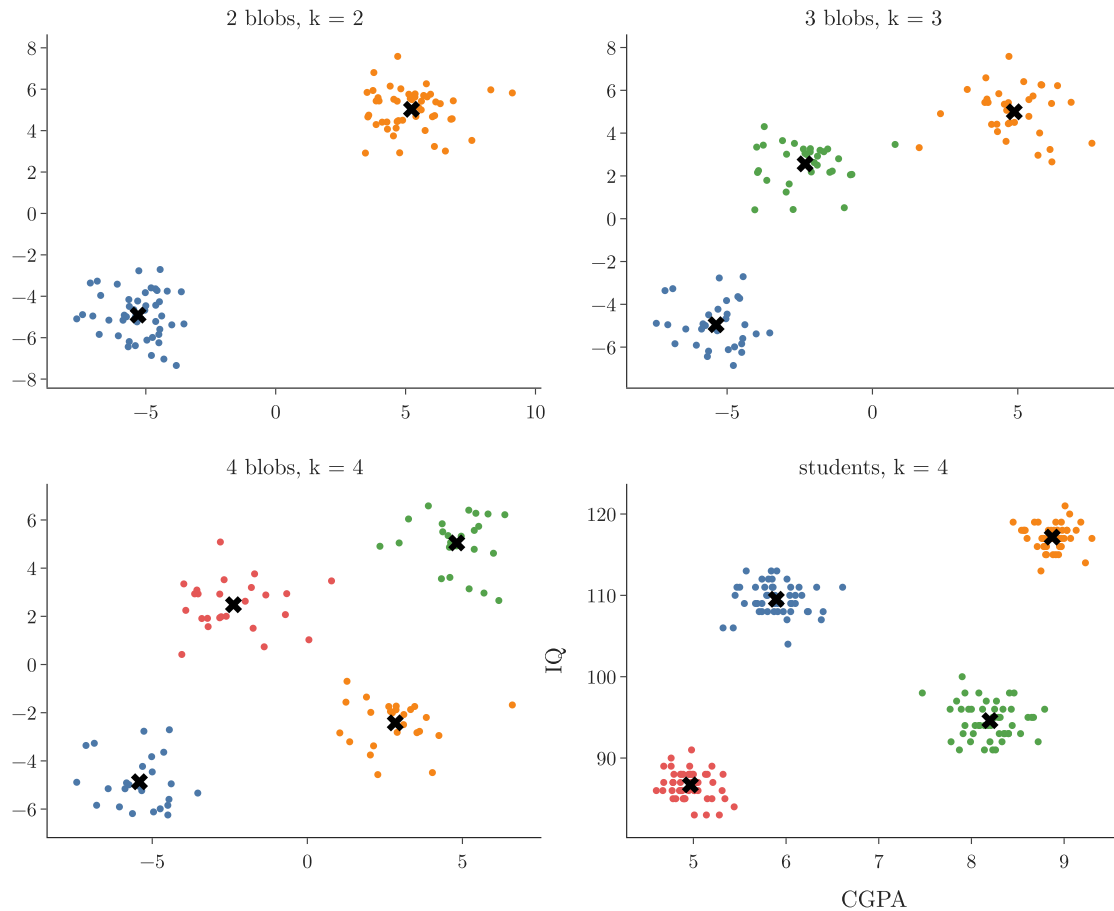


Figure 7: The from-scratch class on 2, 3 and 4 blobs and on the students; crosses are the final centroids

Figure 7 shows all four results: every group found, every centroid in the middle of its group. The loops finished after 2, 5, 3 and 4 rounds. Plotting needs one scatter call per cluster, so the plotting code (not the class) must be extended each time k grows.

Python: The self-check in `kmeans.py`.

```
from sklearn.metrics import adjusted_rand_score
```

```
ours = KMeans(n_clusters=3, random_state=1).fit_predict(X)
```

```
theirs = SkKMeans(n_clusters=3, random_state=0).fit_predict(X)
```

```
assert adjusted_rand_score(ours, theirs) == 1.0
```

Cluster numbers are only names, so we cannot compare them directly. `adjusted_rand_score` is 1.0 when two labelings put exactly the same points together, whatever the numbers.

9. Bad random starts

Key point: If the starting centroids fall badly, k-means gets stuck in poor clusters, a **local optimum** (G-1111); more rounds do not help, but restarting from other random points does.

Sometimes the class returns wrong clusters. Blaming `max_iter` and raising it is tempting, but the round counts say otherwise. Across 30 random starts on the four blobs, k-means never needed more than 10 rounds; at most 10 of the 100 allowed rounds were ever used.

Figure 8 shows the real cause on the students. With one random start (left), two centroids end up in the bottom-left group. That group gets split in two, while two real groups, the top ones, have to share one centroid placed in the empty space between them. After 3 rounds nothing changes any more, so the algorithm stops, stuck. Raising `max_iter` to 500 gives exactly the same result.

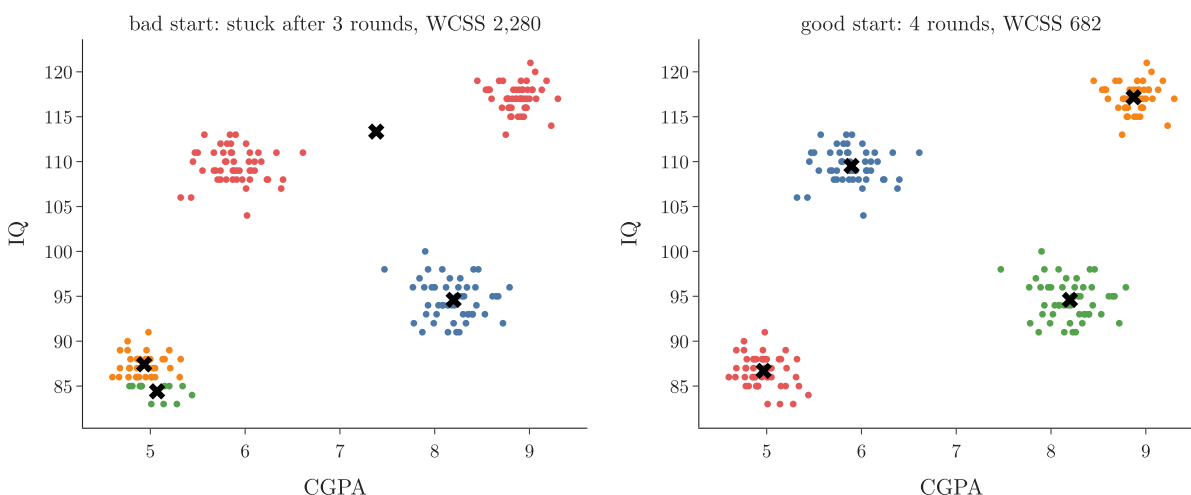


Figure 8: The same students and $k = 4$ from two random starts: stuck in poor clusters (left) and the right clusters (right)

Figure 9 runs both starts round by round. Watch the status in each panel's title. The bad start stops in round 3 and the good start in round 4: in both, the centroids no longer move, so the loop breaks long before `max_iter`. From then on nothing changes in either panel, and the bad run stays at WCSS 2,280. Extra rounds are never used, so they cannot repair a bad start.

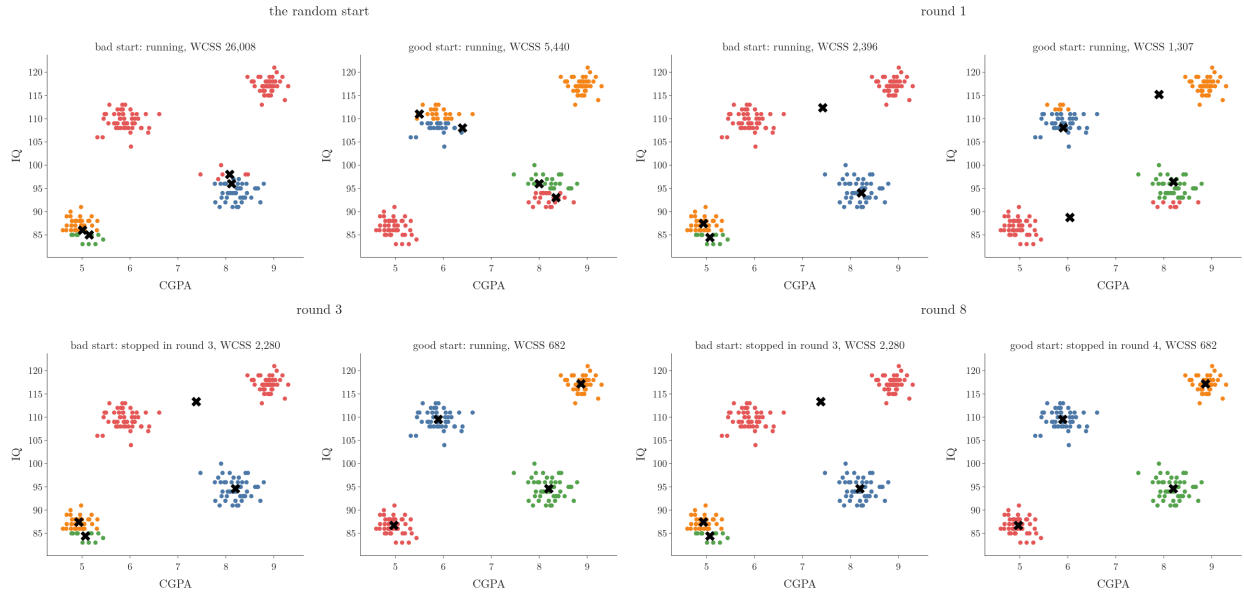


Figure 9: The bad start (left) and the good start (right) on the students, one assign-and-move round per frame, with the WCSS after each round. The bad run stops in round 3 at WCSS 2,280, the good run in round 4 at WCSS 682

The two runs can be compared by their WCSS (the [k-means Note](#), section 5.1): 2,280 for the bad start and 682 for the good one. Lower is better, so the fix is simple (ESL §14.3.6):

1. run k-means from several random starts;
2. keep the run with the lowest WCSS.

In the Notebook, the best of 10 starts is the good clustering (WCSS 682). scikit-learn's `n_init` (the [k-means in Python Note](#), section 5) does exactly this job, and its `k-means++` start spreads the first centroids out.

Extra: The class also stores the WCSS of its result as `inertia_`, like scikit-learn:

```
self.inertia_ = ((X - self.centroids[cluster_group]) ** 2).sum()
```

`self.centroids[cluster_group]` repeats, for every row, the centroid of its own cluster; subtracting, squaring and summing gives the WCSS. With it, the elbow method takes one loop over k . Taking the best of 10 starts for each k , the students give WCSS 29,958, 4,184, 2,363, 682, 514, ...: the elbow is at $k = 4$, as with scikit-learn.

10. Summary

Method	Step	What it does
<code>__init__</code>	1	Stores <code>k</code> (<code>n_clusters</code>) and the round limit (<code>max_iter</code>)
<code>fit_predict</code>	2	Picks <code>k</code> random rows as the first centroids, then runs the loop
<code>assign_clusters</code>	3	Distance from every row to every centroid; keeps the index of the nearest
<code>move_centroids</code>	4	Each centroid becomes the column means of its rows
check in <code>fit_predict</code>	5	Stops when the centroids no longer move

- The Euclidean distance for any number of features: `np.sqrt(np.dot(a - b, a - b))`.
- `X[cluster_group == k].mean(axis=0)` is the new centroid of cluster `k`.
- On the four blobs, k-means stopped within 10 rounds from all 30 starts; the wrong result on the students came from a bad random start, not from too few rounds.
- Fix bad starts by restarting several times and keeping the lowest WCSS.

11. Sources

Built from

- CampusX, “K-Means Clustering Algorithm From Scratch In Python | ML Algorithms From Scratch”, YouTube, <https://www.youtube.com/watch?v=MFraC1JObUo>
- StatQuest with Josh Starmer, “StatQuest: K-means clustering”, YouTube, <https://www.youtube.com/watch?v=4b5d3muPQmA> (a bad start is fixed by rerunning from new starts and keeping the lowest total variation, section 9)

Other references

- ESL: Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009, §14.3.6 (k-means; restarting from several random starts and keeping the lowest objective).
- scikit-learn source, `_k_means_common.pyx`: the function `_relocate_empty_clusters_dense` in `sklearn/cluster/_k_means_common.pyx`, scikit-learn 1.9.

12. Key terms

Term	Meaning
Constructor (<code>__init__</code>)	The method that runs when an object is created and stores its settings
<code>max_iter</code> (G-113)	The largest number of assign-and-move rounds k-means may run
<code>np.argmin</code>	NumPy function returning the position of the smallest value
<code>mean(axis=0)</code>	The mean of each column of an array
Local optimum (k-means)	A clustering where k-means has stopped but a better one exists, caused by a bad start
Adjusted Rand score (G-175)	A number that is 1.0 when two labelings group the points identically, whatever the label numbers